

Abhinaya Chittedi

Sr. DevOps Engineer (Azure / AWS)

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Professional Summary:

Accumulated 10+ years of professional IT experience encompassing various domains such as Software Development, DevOps, Testing, Configuration Management, Build and Deployment Automation, Source Code Repository Management, and Environment Management.

- Proficiently managed application software implementation and maintenance across diverse Web and Cloud/Server Environments, including Amazon Web Service (AWS) and Azure.
- Developed an ETL pipeline using AWS Glue to automate the extraction, transformation, and loading of data from various data sources such as S3 and relational databases.
- Demonstrated expertise in DevOps practices, Release Engineering, Configuration Management, and Cloud Infrastructure Automation tools. Proficient with Git, Bitbucket, Maven, Jenkins, TeamCity, Chef, Ansible, Terraform, Docker, Kubernetes and ELK.
- Leveraged Terraform to automate the provisioning of VPCs, ELBs, Security Groups, SQS queues, and S3 buckets, transitioning towards infrastructure as code (IaC).
- Set up and configured OpenSearch for efficient indexing and querying, implementing custom queries and index lifecycle management (ILM) to optimize performance and storage costs.
- Extensive experience in provisioning IAAS and PAAS concepts in cloud computing platforms such as AWS, Azure, and Google Cloud Platform, while adhering to Agile, Scrum, and Waterfall methodologies.
- Designed a DynamoDB schema with efficient indexing, automated bulk data processing using **boto3**, and implemented real-time event-driven processing via DynamoDB Streams.
- Implemented DevOps practices for Micro Services and Docker-based clusters, employing Kubernetes and Spinnaker as orchestrators for cloud platform application development, with a focus on Spring Boot and Microservices.
- Built & maintained Docker container clusters, managed AWS environments, & utilized technologies like Linux, Bash, GIT & Docker. Employed Kubernetes pods & Docker to create a runtime environment for the CI/CD system, facilitating efficient build, testing, & deployment.
- Built a graph data model in AWS Neptune and wrote Cypher and Gremlin queries for advanced relationship analysis, including shortest path calculations and community detection.
- Proficiently configured CI (Continuous Integration) pipelines, with expertise in Jenkins and TeamCity server setup, script creation (Bash shell, Perl, Python), and VM configuration (Windows/Linux). Additionally, generated Infrastructure reports using PowerShell scripts.
- Automated and orchestrated the entire data pipeline using AWS Glue, Step Functions, and Lambda, ensuring seamless data processing and real-time monitoring with CloudWatch.
- Extensive experience in troubleshooting, configuring, and deploying enterprise applications on WebSphere, WebLogic, Tomcat, and JBoss, while conducting SCM verifications and audits to ensure SOX compliance.
- Proficient in addressing database performance-related issues in MySQL and Oracle databases.
- Demonstrated hands-on expertise with configuration management tools such as Chef, Puppet, and Ansible, along with a strong commitment to continuous learning and tool adoption.
- Capable of efficiently handling multiple tasks, working both collaboratively in a team and independently to meet project deadlines.
- Implemented strong security practices using IAM roles, encryption, and auto-scaling mechanisms while optimizing performance in OpenSearch, DynamoDB, and Spark jobs for scalability and efficiency.
- Proficiently submitted implementation requests to various stakeholder groups involved in the release process.
- Strong background in system administration, including installation, upgrading, patch management, migration, configuration, troubleshooting, security, backup, and disaster recovery in environments spanning AWS, Azure, AKS, and OpenStack.

- Utilized OpenShift for container orchestration alongside Kubernetes, focusing on container storage, automation, and multi-tenancy enhancement within the container platform.
- Applied hands-on experience in designing modules, creating playbooks, and configuring inventories in Ansible.
- Expertise in server monitoring, capacity planning, and application monitoring, employing tools such as Splunk, GitHub/GitLab, AWS CloudWatch, and Kibana.
- Involved in developing Java web-based interfaces using HTML5, CSS3, JQuery, XML and JSON.
- Good understanding of the principles and best practices of Software Configuration Management (SCM) in Agile, scrum, and Waterfall methodologies.
- Managed different infrastructure resources, like physical machines, VMs and even Docker containers using Terraform.
- Worked with Ansible (automation tool) to automate the process of deploying/testing the new build in each environment, setting up a new node and configuring machines/servers.
- Written Cloud formation templates and deployed AWS resources using it.
- Excellent written and verbal communication skills, strong organizational skills, and a hard-working team player.

Certifications:

- Certified Azure Administrator Associate.
- Certified AWS Developer Associate.
- Certified Kubernetes Administrator.

Technical Skills:

Cloud Platforms	Microsoft Azure, Aws Cloud, Google Cloud Platform
AWS Services	RDS, EC2, VPC, IAM, Lambda Function, Cloud Formation, EBS, S3, ELB, Auto Scaling, Cloud Trail, SQS, SNS, SWF, Cloud Watch, WAF, ECR, EKS.
Azure Services	App Services, Key vault, function app, Blob storage, Azure Active Directory (Azure AD), Service Bus, Azure Container Registry (ACR) and Azure Kubernetes service (AKS), Azure SQL, Azure Cosmos DB.
CI/CD	Jenkins, Azure DevOps Pipelines, CircleCI, GitLab.
Version Control Tools	GIT, Bit Bucket.
Configuration & Automation Tools	Ansible, Puppet, Chef
Container Platforms	Docker, Kubernetes, Red Hat Open Shift.
Monitoring Tools	Nagios, Splunk, Prometheus, Grafana, Datadog, New Relic.
Artifactory	Jfrog and Nexus
Web Servers	Apache Tomcat, WebSphere, Nginx
Documentation	Confluence
Operating Systems	Microsoft Windows XP/ 2000, Linux, UNIX.
Tracking Tools	Jira, ServiceNow, Azure Board.
Code Scanning	Sonar Qube, Jfrog X ray, ECR Inspector
Databases	RDS, Cosmos DB, My SQL, Aurora DB, Mongo DB.
Logging	Cloud Watch, Cloud Trail, Azure App Insights, Azure Monitor, ELK, Fluent
Languages	Python, Shell scripting, Golang.

PROJECT EXPERIENCE:

Client	Siemens – Atlanta, GA.	Jul 2023 to present
Role	Sr. DevOps Engineer / AWS	
<u>Responsibilities</u>	▪ Designed and administered AWS S3 buckets, implementing policies and utilizing S3 and Glacier for robust storage and backup solutions within the AWS ecosystem. ▪ Leveraged AWS Identity and Access Management (IAM) to enable fine-grained permissions management and resource allocation, proficiently managing user roles and permissions.	

	▪ Architected and executed end-to-end server build management, monitoring, and deployment automation using Ansible, fostering efficiency and consistency. ▪ Established proactive monitoring mechanisms through AWS CloudWatch alarms, closely tracking server performance metrics such as CPU utilization and disk usage, and initiating data-driven optimizations. ▪ Optimized the performance of PySpark jobs by fine-tuning configurations like memory settings and partition sizes, significantly improving processing times. ▪ Implemented shard management and replica optimization in OpenSearch to handle large-scale search queries efficiently. ▪ Ensured DynamoDB scaling by using auto-scaling mechanisms and optimizing table design for read-heavy workloads. ▪ Configured AWS Multi-Factor Authentication (MFA) via IAM, enforcing two-factor authentication using Google Authenticator and AWS Virtual MFA for enhanced security. ▪ Implemented robust security measures encompassing security groups, network ACLs, Internet Gateways, and Elastic IPs, fortifying the organization's presence in the AWS public cloud. ▪ Implemented encryption for data at rest and in transit using AWS KMS, ensuring compliance with best practices for security. ▪ Managed DNS infrastructure utilizing Amazon Route 53, overseeing DNS zone administration and associating public DNS names with Elastic Load Balancer IPs. ▪ Developed Ansible Playbooks with PythonSSH wrappers for streamlined AWS Node configuration management and executed playbook testing on AWS instances using Python. Proficiently employed Ansible Tower for enterprise-scale deployments. ▪ Designed AWS CloudFormation templates to dynamically create custom-sized VPCs, subnets, and NAT gateways, facilitating the seamless deployment of web applications and database templates. ▪ Deployed Ansible and Ansible Playbooks to orchestrate configuration management across the existing infrastructure, ensuring consistency and scalability. ▪ Collaborated closely with development teams to establish and enforce branching and naming conventions in GIT source control. ▪ Hands-on experience configuring AWS network architecture, including VPCs, subnets, Internet gateways, NAT gateways, & route tables, reinforcing network security & performance. ▪ Assumed responsibility for systems and network security, maintaining optimal performance, and setting up comprehensive monitoring via CloudWatch . ▪ Distinguished experience in crafting Splunk applications and Enterprise Security solutions to proactively identify and mitigate emerging security threats through continuous monitoring, alerting, and advanced analytics. ▪ Expertly utilized Splunk to parse, index, and analyze extensive volumes of log data from diverse sources, including servers, applications, and network devices. ▪ Demonstrated extensive experience in architecting and configuring secure cloud VPCs, intricately configuring private and public networks via subnets within AWS. ▪ Designed and constructed comprehensive CI/CD pipelines for projects, encompassing builds, testing, static code analysis, Maven versioning, and deployments, laying the foundation for department-wide application development. ▪ Implemented robust security measures within Jenkins, encompassing user access control and plugin management, bolstering platform stability. ▪ Leveraged Jenkins plugins for seamless cloud integration, enabling dynamic resource scaling for CI/CD processes across AWS and Azure platforms. ▪ Skillfully designed, deployed, and orchestrated containerized applications leveraging Docker, while ensuring high availability and scalability with Kubernetes.
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	- Utilized Terraform for Infrastructure as Code (IAC), proficiently scripting EC2 instances, Elastic Load Balancers, and S3 buckets, ensuring infrastructure consistency and reliability. - Effectively managed AWS infrastructure provisioning via CloudFormation and Terraform, automating resource lifecycle management and maintaining resource state in Terraform state files.
Environment	AWS, Terraform, Chef, Ansible, YAML, Docker, Kubernetes, Openshift, EKS, Azure CLI, Jenkins, AKS, Git, Maven, Splunk, Nagios, Python Scripts, Kickstart, Java/J2EE, Linux

Client	SMBC – Jersey City, NJ. May 2022 to Jun 2023
Role	Sr. DevOps Engineer / Azure
<u>Responsibilities</u>	- Orchestrated Slot Blue-Green Deployments for Azure App Services. - Containerized Jenkins pipelines during the migration to Azure DevOps. - Leveraged Azure services, including Azure DevOps, Azure Kubernetes Services (AKS), Container Registry, Cosmos DB, Grafana, Azure Pipelines, Monitor, and RBAC for microservices development, application migration, and management. - Deployed Azure IaaS virtual machines (VMs) and Cloud services (PaaS role instances) into secure VNets and subnets. - Worked on Server less services (Azure Functions) created, and configured HTTP Triggers in Azure Functions with Application insights for monitoring and performing load testing on the Applications using the VSTS. - Implemented end-to-end orchestration of the ETL pipeline using AWS Glue, Step Functions, and Lambda, enabling automated, scheduled, and reliable data processing. - Integrated monitoring and logging through AWS CloudWatch to track job status, performance metrics, and handle errors gracefully via custom alerting systems. - Written Templates for Azure infrastructure as a code using Terraform to build staging and production environments. Integrated Azure Log Analytics with Azure Virtual machines for monitoring the log files, store them and track metrics - Built a graph-based data model using AWS Neptune to model and analyze relationships between entities, such as customer interactions in a recommendation system. - Developed Cypher and Gremlin queries to explore graph relationships, including shortest path algorithms and community detection, to extract meaningful insights from the graph data. - Optimized query performance for large-scale graph traversals by implementing indexing and efficient query patterns. - Implemented Ansible as a configuration management system for web applications, environment configurations, users, mount points, and package management. - Engineered pre-commit hooks for GitHub/GitLab integration with Amazon EKS, Jenkins, and JIRA, complemented by post-commit hooks for automated continuous builds. - Automated build, test & deployment processes by tightly integrating Git with CI/CD pipelines. - Orchestrated middleware artifact deployments through Jenkins job configurations. - Configured intricate cloud infrastructures, featuring Virtual Networks, Virtual Machines, Azure Active Directory, Load Balancers, Azure SQL, and Cloud Services. - Architected, deployed, and fine-tuned hybrid cloud components such as Azure AD, ADFS, SSO, and VPN Gateways to ensure uninterrupted business operations. - Spearheaded the development and maintenance of Splunk-based monitoring and alerting systems, resulting in a 40% reduction in mean time to resolution (MTTR) and a 25% increase in system availability. - Employed advanced Splunk configurations to filter and manage log data effectively, including the utilization of Splunk as a SIEM solution for real-time security monitoring. - Managed and maintained multiple Web Logic Servers, guaranteeing load balancing, high availability, and seamless failover capabilities.

	▪ Excelled in Docker containerization, crafting Docker files, orchestrating automated builds, and streamlining multi-container provisioning with Docker Compose. Further streamlined container management via Azure Container Registry. ▪ Pioneered cluster management solutions using Docker and AKS to facilitate local Kubernetes deployments, including the creation of self-hosted Kubernetes clusters through Jenkins & Git. ▪ Showcased Kubernetes proficiency through the creation of pods, controllers, services, deployments, health checks, and ingress configurations using YAML files. ▪ Utilized Terraform templates for Azure infrastructure provisioning, featuring virtual machines, scale sets, Azure Site Recovery, and Azure Backup. Further enhanced infrastructure control through Log Analytics data collection. ▪ Demonstrated expertise in Ansible, creating modular components, composing intricate playbooks, and meticulously configuring inventory. ▪ Spearheaded the implementation of an ELK stack for centralized logging, driving real-time log analysis, anomaly detection & swift issue resolution, culminating in a 30% reduction in MTTR. ▪ Proficiently managed Private Cloud and Hybrid Cloud configurations, spanning Windows Azure, SQL Azure, Azure Web, and Database deployments. ▪ Developed PowerShell scripts for orchestration and harnessed Dynatrace for monitoring and support.
Environment	Azure, Terraform, Chef, Ansible, Docker, Kubernetes, Jenkins, Git, Maven, Splunk, Nagios, Python Scripts, Kickstart, Java/J2EE, Linux.

Client	State of NE – Lincoln, NE Feb 2021 to Apr 2022
Role	DevOps Engineer
<u>Responsibilities</u>	▪ Worked on many different concepts of Chef like Roles, Environments, Data Bags, Knife, and Chef Server Admin/Organizations. ▪ Written Chef Recipes to automate build/deployment process and data bags in Chef for better environment management. ▪ Tested Chef Cookbook modifications on cloud instances in AWS and using Test Kitchen and Chef Spec ▪ Designed the DynamoDB schema to optimize for high throughput and low-latency data access, considering partition keys and secondary indexes for querying flexibility. ▪ Developed scripts using boto3 to perform bulk data inserts, updates, and complex queries, ensuring data consistency and high availability. ▪ Utilized DynamoDB Streams to trigger Lambda functions for event-driven data processing and real-time analytics. ▪ Written Chef Cookbooks for various DB configurations to modularize and optimize product configuration, converting production support scripts to Chef Recipes and AWS server provisioning using Chef Recipes. ▪ Used SonarQube for continuous inspection of code quality and to perform automatic reviews of code to detect bugs. ▪ Automated Nagios alerts and email notifications using Python script and executed them through Chef. ▪ Installed, configured and maintained web servers like Apache Web Server and WebSphere Application Server on Red Hat Linux. ▪ Experience with Red Hat Linux kernel, memory upgrades and swaps area. ▪ Red Hat Linux Kickstart. Installation Sun Solaris Jumpstart Installation. Configuring DNS, DHCP, NIS, NFS in Sun Solaris 8/9 & other Network Services. ▪ Used multiple EC2 instances at the same time and provided highly durable and available data by using S3 data store, versioning, lifecycle policies, & create AMIs for mission-critical production server's backup. ▪ Provided highly durable and available data by using S3 data store, versioning, lifecycle policies.

	- Performed the automation deployments using AWS by creating the IAMs and used the code pipeline plugin to integrate Jenkins with AWS and created the EC2 instances to provide the virtual servers. - Worked with Knife command line tool for creating Recipes, Cookbooks and worked with chef supermarket. - Implemented Docker-Maven plugin and Maven POM to build Docker Images for all microservices and later used Docker file to build the Docker Images from the java jar files. - Used Git for source code version control and integrated with Jenkins for CI/CD pipeline, code quality tracking and user management with build tools Maven and Ant and written Maven pom.xml build script. - Worked on NoSQL database MongoDB to replica setup and sharing. - Also experienced in managing replica set. - Installed, Configured, and Managed Monitoring Tools such as Nagios for Resource Monitoring/ Network Monitoring.
Environment	AWS (EC Route, EBS, Security Group, Auto Scaling, and RDS), GIT, Chef, Docker, SonarQube, Maven, Jenkins, ANT, Python, Nagios.

Client	Ally Financial - Charlotte, NC.		Nov 2019 to Jan 2021
Role	DevOps Engineer		
<u>Responsibilities</u>	- Experience in creating Docker containers leveraging existing Linux containers and AMIs in addition to creating Docker containers from scratch. - Worked on creation of custom Docker container images, pushing the images and Docker consoles for managing the application lifecycles. - Utilized key features in Terraform such as Infrastructure as Code (IAC), Change automation, Execution plans. Wrote Terraform templates to deploy infrastructure on cloud with EC2, ELB. - Developed an ETL pipeline using AWS Glue to automate the extraction, transformation, and loading of data from various data sources such as S3 and relational databases. - Utilized PySpark for processing large datasets in parallel, optimizing data cleansing, aggregation, and transformations at scale. - Designed custom AWS Glue jobs to automate recurring data processing workflows and utilized AWS Glue Crawlers to handle dynamic schema detection and cataloging. - Configured AWS IAM and security groups in Public and Private subnets in VPC. - Worked on designing and deploying applications using AWS services like EC2, Route53, S3, RDS, DynamoDB, SNS, SQS and ELK. Focused on high-availability, fault tolerance and auto scaling in AWS Cloud Formation. - Involved in building configuration management for Microservices using Docker, Kubernetes and Ansible. Used Kubernetes to setup load balancing and service selection. - Worked on managing Private cloud environment using Ansible and Enhanced the automation to assist configuration management using Ansible based YAML scripts. - Worked on Docker container snapshots, attaching to a running container, removing images, managing Directory structures, and managing containers. - Managed local deployments in Kubernetes, created local cluster and deploying application containers. created private cloud using Kubernetes that supports Dev, Test and Prod environments. - Configured Jenkins for doing the build in all the non-production and production environments. - Used Git with Jenkins to integrate to automate the code checkout process. - Configured Jenkins for doing the build in all non-production environments and most of the production environments.		
Environment	AWS, Docker, Kubernetes, Jenkins, YAML, Ansible, Terraform, GIT, MAVEN, Jenkins, AWS Cloud, Docker, PowerShell.		

Client	PeopleNet – Hopkins, MN.	Jun 2018 to Oct 2019
Role	Build Release Engineer. / DevOps Engineer	
<u>Responsibilities</u>	▪ Defined and Implemented CI and Release Management Processes, Policies, and Procedure. ▪ Automated the release pipeline to achieve zero touch deployments using Jenkins, SVN, Nexus, and Nolio. ▪ Implemented roles and groups for users and resources using AWS Identity Access Management (IAM) and managed network security using Security Groups and IAM. ▪ Implemented automations tools Chef, Puppet, and DevOps. ▪ Administer and Maintain Source Code Repositories including GIT, Subversion SVC. ▪ Automated MySQL container deployment in Docker and monitoring of these containers using Nagios. ▪ Involved in developing interactive web applications using dynamic scripting language like PHP, and Python Perl. ▪ Deploying Java Enterprise applications to Apache Web Server, JBoss Application server. ▪ Responsible for building and automating monitoring and performance management of all production environments, which includes servers, network equipment, databases and proprietary software applications. ▪ Installed, configured, and provided support for Tivoli Monitoring software across various OS platforms like RHEL, AIX and Solaris. ▪ Implementing new projects builds framework using Bamboo & maven as build framework tools. ▪ Implemented AWS High-Availability using AWS Elastic Load Balancing (ELB), which performed a balance across instances in multiple Availability Zones. ▪ Worked with SVN and GIT version controlling ▪ Developed Scripts for AWS Orchestration using EC2, EBS, S3, Cloud Formation and IAM. ▪ Develop/capture/document architectural best practices for building systems on AWS. ▪ Goal is to build a HA proxy Infrastructure so that two docker containers are running behind Application load balancer. ▪ Implement, maintain, and support reliable, timely and reproducible builds for project teams. ▪ Maintained build related scripts developed in shell for Maven builds. ▪ Created and modified build configuration files including POM.xml. ▪ Utilized Perl and Python for GIT for creating repository hooks ▪ Used Maven dependency management system to deploy snapshot and release artefacts to Nexus to share artefacts across projects and environments.	
Environment	AWS, Docker, SVN, Jenkins, Nexus, GIT, ANT, MAVEN, Perl, Puppet, Ansible, Python Scripts, Shell Scripts, Sonar, Red Hat Kickstart server, Splunk.	

Client	Aveva, Hyderabad, India.	Jul 2013 to Sep 2017
Role	Linux Administrator	
<u>Responsibilities</u>	▪ Installation, configuration, backup, recovery, maintenance and support of RedHat Linux and Solaris. ▪ Troubleshooting Day-to-Day issues with various Servers on different platforms. ▪ File system Administration, Setting up Disk Quota, solutions on SAN/NAS storage. ▪ Configuration and administration of Clustered servers on SUSE Linux environment. ▪ Configuring backups on newly built servers and monitoring failed backups. ▪ Install and configure Ubuntu, Centos on remote servers and Desktop servers. ▪ Installation of patches and packages using RPM and YUM in Red hat Linux.	

	▪ Installed and configured Apache/Tomcat web server. ▪ Provided 24x7 on-call supports in debugging and fixing issues related to Linux, Solaris. ▪ HP-UX Installation/Maintenance of Hardware/Software in Production, Development. ▪ Responsible for maintenance of source control systems Subversion and GIT. ▪ Created repositories according to the structure required with branches, tags, and trunks. ▪ Created hooks and managed permissions on the branches for GIT. ▪ Responsible for designing and deploying best SCM processes and procedures. ▪ Installed and configured SSH server on Red hat/Cent OS Linux environments. ▪ Created and maintained several build definitions and publish profiles to handle automated builds in an Agile/SCRUM environment. ▪ Attended sprint planning sessions and daily sprint stand-up meetings. Configured application servers (JBOSS) to deploy the code. ▪ Used Apache Tomcat as application server for deploying the artifacts. ▪ Installed, administered, and configured Jenkins/Bamboo Continuous Integration tool. ▪ Used Ant, Maven to perform daily and weekly Software Builds. ▪ Helped developers and other project teams to set views and environments. ▪ Replicated the Jenkins build server to a test VM using Packer, Virtual Box, Vagrant, Chef, Perl brew and Server spec.
Environment	Linux, UNIX, RHEL, CentOS, Ubuntu, NIS, DHCP, FTP, Telnet, Nagios, Kickstart, SSH, VM Sphere, VMWare Virtual Box, RPM and YUM, Logical Volume Manager (LVM), Internet Protocol Suite (TCP/IP), Tomcat Apache, Firewalls, VMware, Unix.